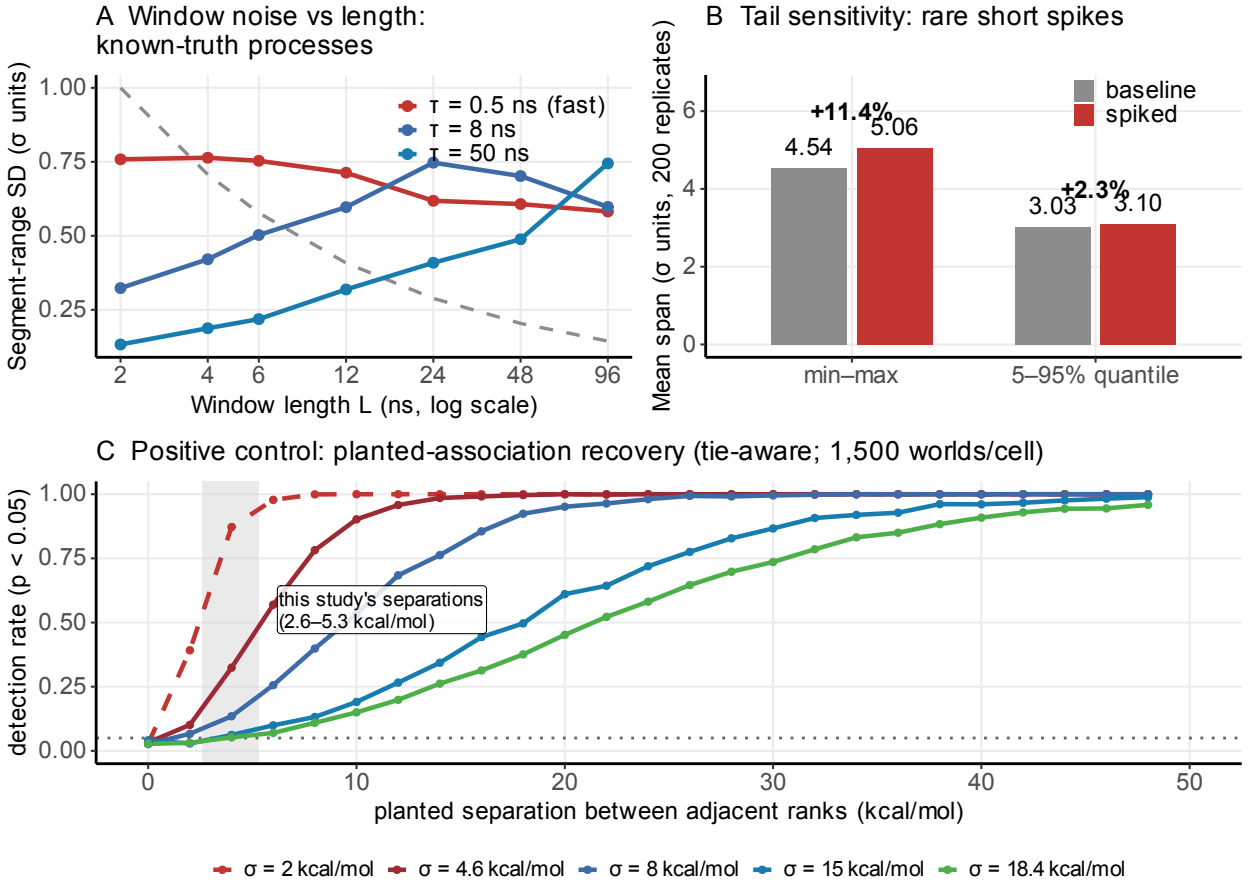
A Window noise vs length:
known-truth processes
Segment-range SD (σ units)
1.00
0.75
0.50
0.25
τ = 0.5 ns (fast)
τ = 8 ns
τ = 50 ns
2
4
6
12
24
48
96
Window length L (ns, log scale)
B Tail sensitivity: rare short spikes
Mean span (σ units, 200 replicates)
6
4
2
0
baseline
spiked
+11.4%
4.54
5.06
+2.3%
3.03
3.10
min–max
5–95% quantile
C Positive control: planted-association recovery (tie-aware; 1,500 worlds/cell)
detection rate (p < 0.05)
1.00
0.75
0.50
0.25
0.00
this study's separations
(2.6–5.3 kcal/mol)
0
10
20
30
40
50
planted separation between adjacent ranks (kcal/mol)
σ = 2 kcal/mol
σ = 4.6 kcal/mol
σ = 8 kcal/mol
σ = 15 kcal/mol
σ = 18.4 kcal/mol